DevKnowledge AI Platform

React Frontend Architecture, Security Blueprint & AWS Infrastructure Guide

1. EXECUTIVE ARCHITECTURE SUMMARY

this technical reference details the component layout, environment isolation, secrets management, and cloud deployment pipeline for the DevKnowledge Frontend Application. Built with React (Vite) and Material UI, the app interfaces with AWS CloudFront, S3, API Gateway, and Secrets Manager.

2. PROJECT DIRECTORY STRUCTURE & KEY FILES

```
devknowledge-ui/ ├── .env.development # Dev environment overrides (Git-tracked, non-secre
```

3. KEY MODULE RESPONSIBILITIES

WorkspacePage.jsx

Serves as the main layout frame containing the fixed left sidebar navigation, top app header, user profile dropdown, and standard margin wrapper for workspace pages.

Assistant.jsx

The entry orchestrator for the AI tool. Fetches user projects, handles selection state via MUI dropdowns, and conditionally renders the chat window.

AIChat.jsx

Encapsulates prompt submission, response rendering, streaming state, error handling, and message history tied specifically to the selected `projectId`.

projectService.js

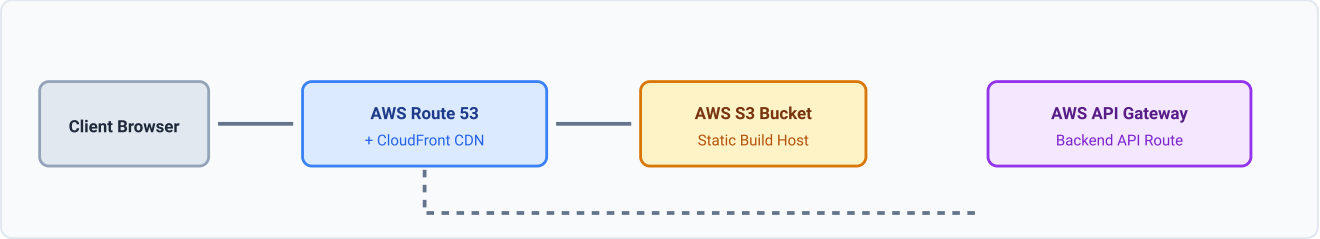
Abstracts HTTP requests using Axios. Handles endpoints for fetching project lists, querying documents, and triggering AI indexing jobs.

4. ENVIRONMENT URLS & ACCESS POINTS

Environment	Status	Frontend URL (AWS CloudFront)	Backend API Endpoint
Production	LIVE	`https://app.devknowledge.io`	`https://api.devknowledge.io/v1`
Staging / Demo	TESTING	`https://demo.devknowledge.io`	`https://api-demo.devknowledge.io/v1`
Development	ACTIVE	`https://dev.devknowledge.io`	`https://api-dev.devknowledge.io/v1`

5. AWS DEPLOYMENT ARCHITECTURE

The React application is compiled into static assets via Vite and hosted on Amazon S3. Traffic is globally distributed via AWS CloudFront CDN with SSL termination. AWS Route 53 routes incoming domain queries.



6. SECRETS & ENVIRONMENT CONFIGURATION STORAGE

Information Type	Storage Location	Security Protocol
Public Config (API Base URL, App Version)	`.env.production`, `.env.development`	Checked into Git repository. Exposed to browser build.
Deployment Keys (AWS Access Key ID, Secret Key)	AWS Secrets Manager / GitHub Secrets	**Never in client code.** Injected directly into GitHub Actions / AWS CodePipeline runtime.
User Auth Tokens (JWT Session Tokens)	Browser `HttpOnly` Cookies / Memory	Protected against XSS attacks; automatically attached via Axios interceptor (`api.js`).
Third-Party API Keys (OpenAI, Anthropic)	AWS Parameter Store (Backend side)	Frontend never calls third-party APIs directly; requests are proxied via Backend API Gateway.